



University of Al-Ameed / College of Medicine
Department of Physiology and Medical Physics



Physiology of the cardiovascular system

Lecture -1-

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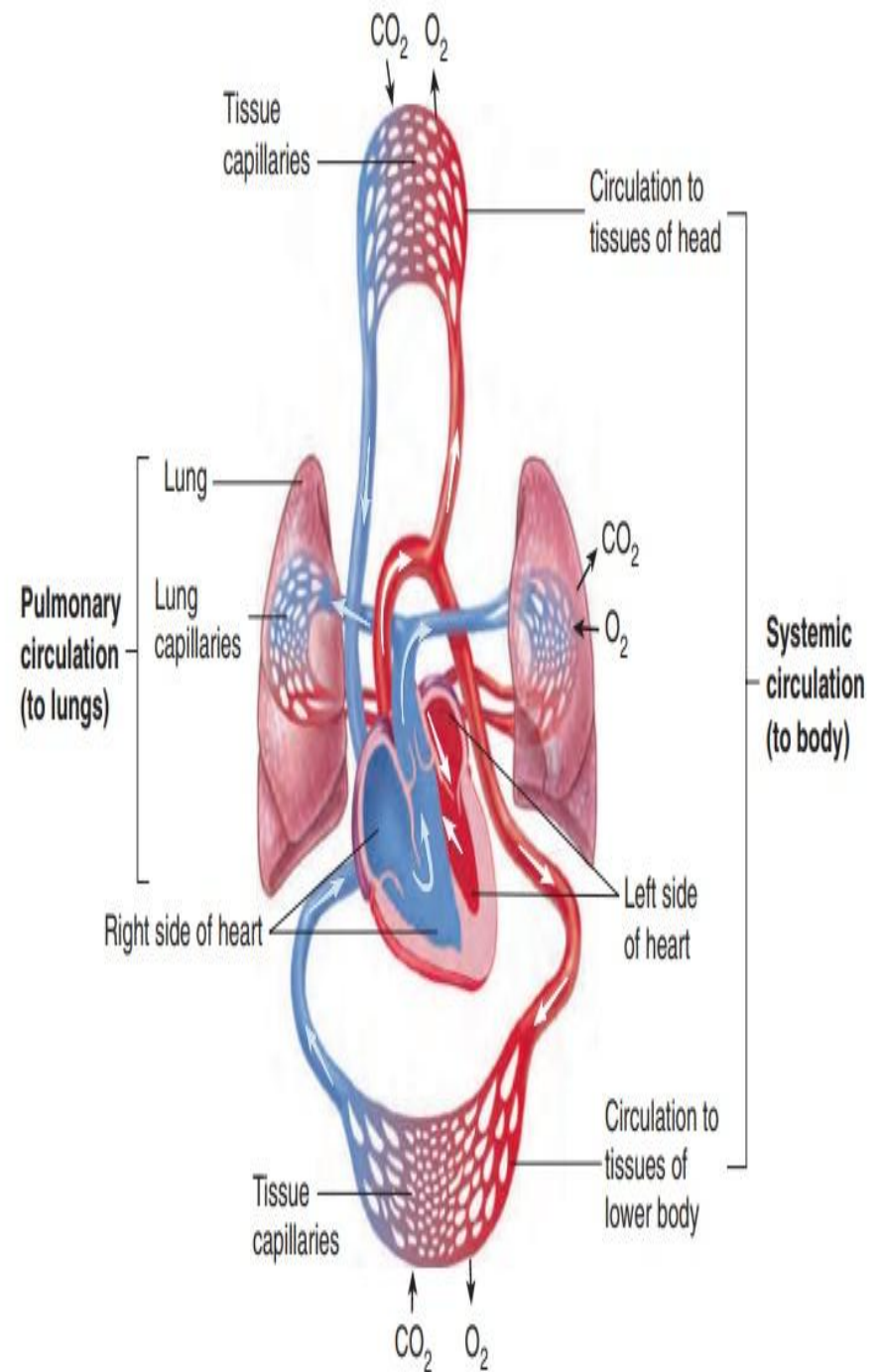
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Learning Objectives

1. General structure of the heart and its valves
2. General structure and features of cardiac muscle
3. General feature of pulmonary and systemic circulation
4. Structure and features of the cardiac conductive system

Functions of the Heart

1. Producing blood pressure
2. Directing blood flow by distinguishing between pulmonary and systemic circulations
3. Maintaining unidirectional blood flow through valves
4. Controlling blood distribution
5. Adjusting contraction rate and strength to align blood delivery with varying metabolic demands



The cardiovascular system consists of two main circuits:

Pulmonary circuit

Responsible for transporting blood between the heart and the lungs.

Systemic circuit

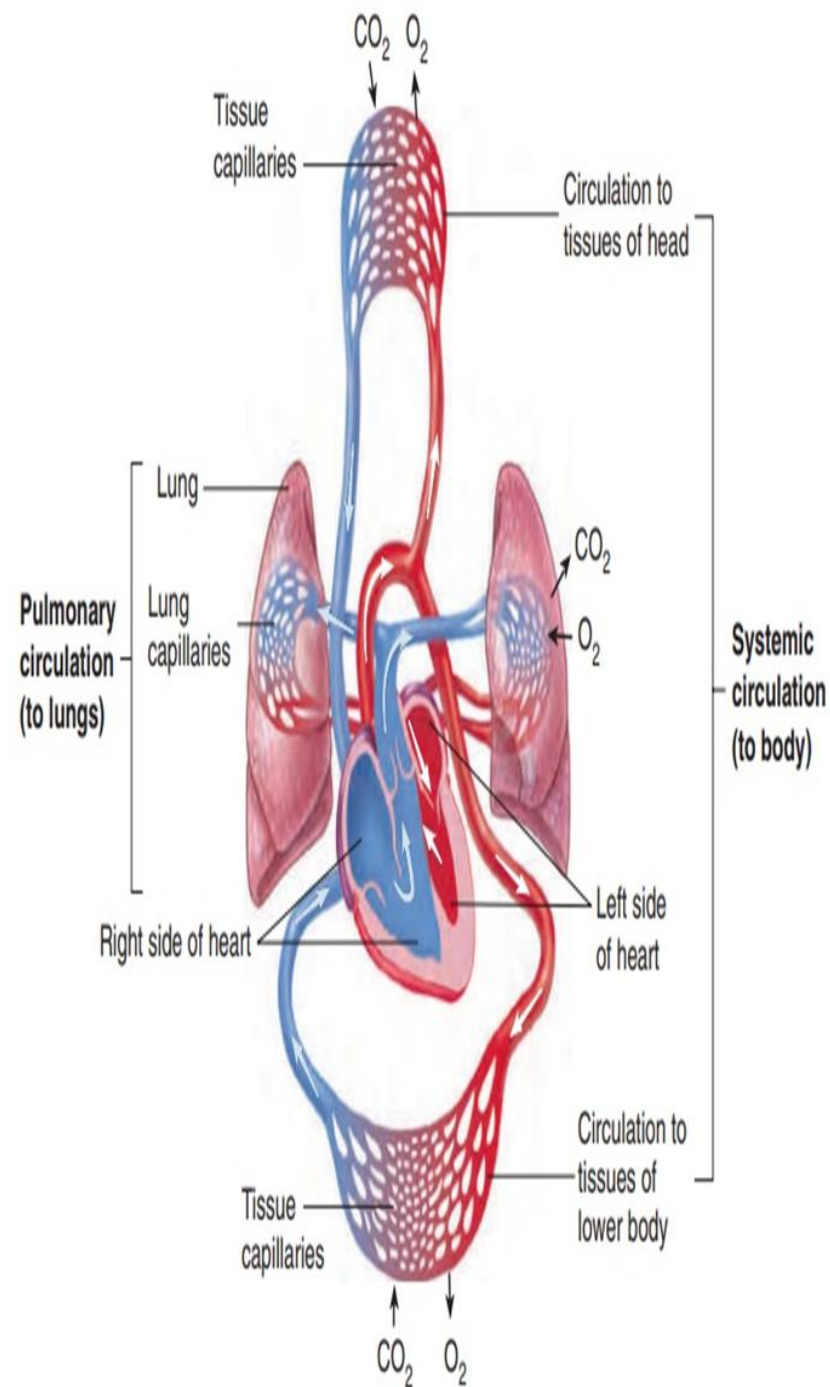
Responsible for circulating blood between the heart and the rest of the body.

Blood flows through these circuits via various vessels:

Arteries transport blood away from the heart.

Veins return blood to the heart.

Capillaries facilitate the exchange of gases, nutrients, and waste products.



Blood Flow through the Heart

❖ Right ventricle (RV)

receives blood from RA

pumps to lungs via Pulmonary Trunk (PT)

❖ Pulmonary Trunk (PT) -

from RV branches into the pulmonary arteries (PA)

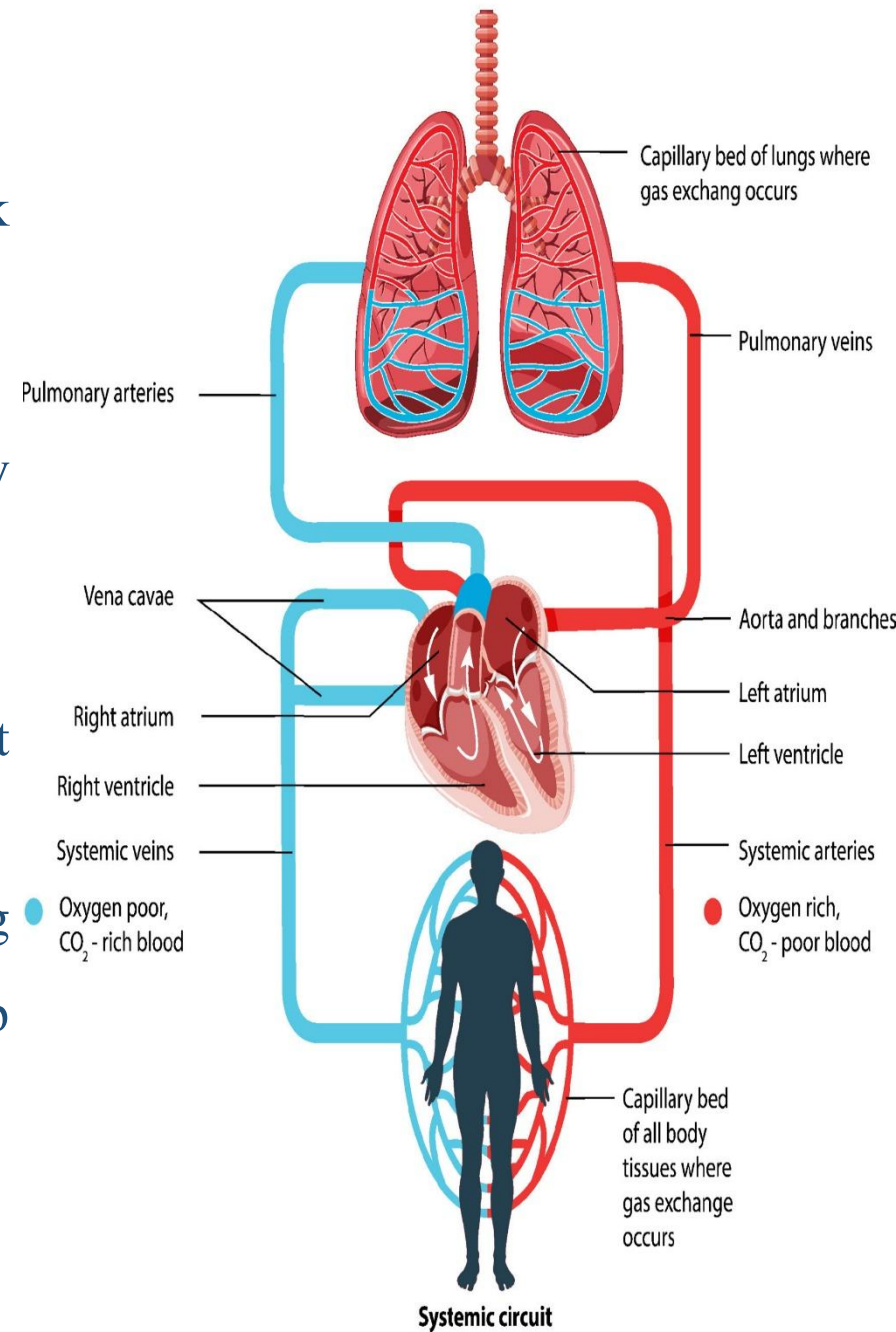
❖ Pulmonary arteries

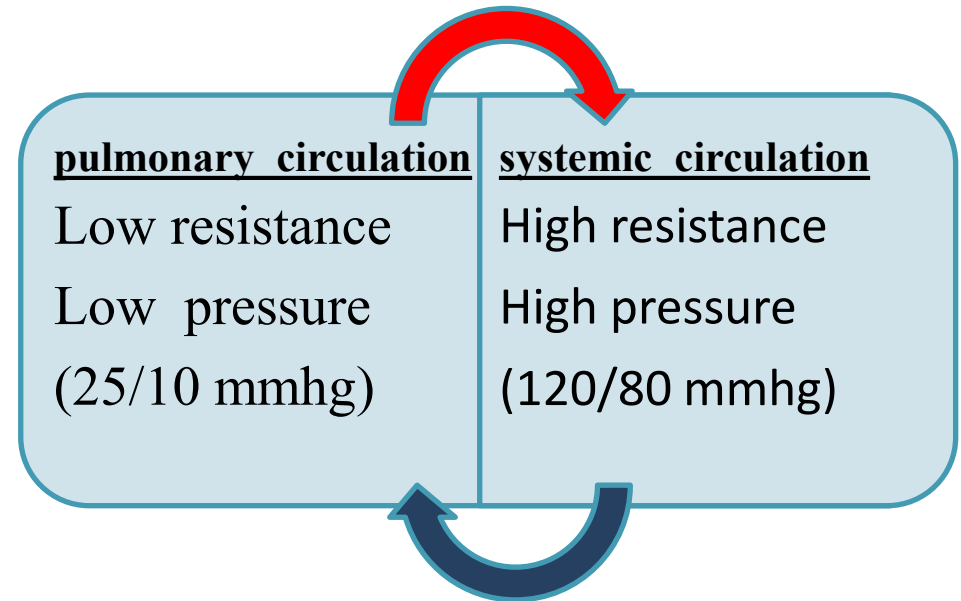
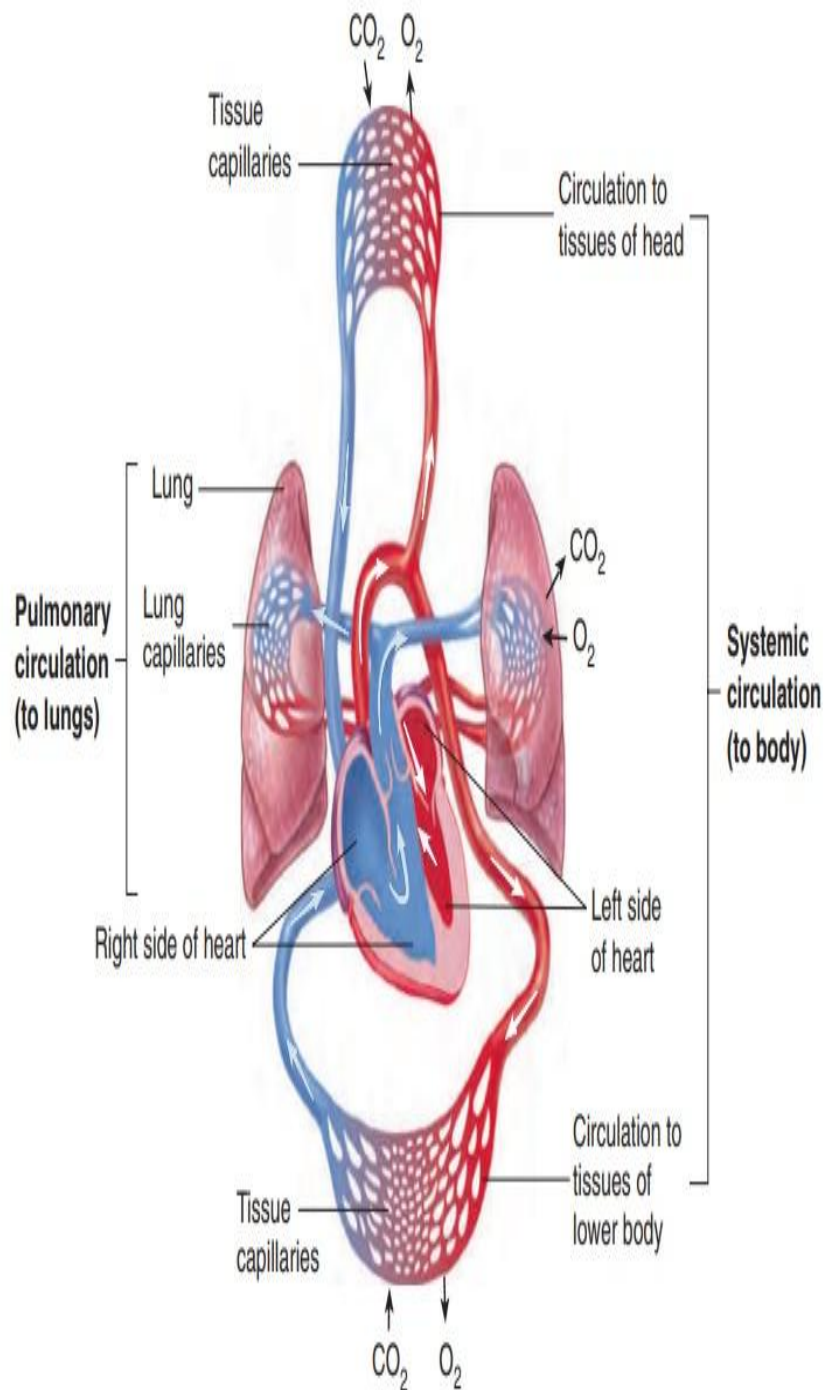
- deoxygenated blood from the heart to the lungs for gas exchange
- right and left branches for each lung
- blood gives up CO₂ and picks up O₂ in the lungs

❖ Pulmonary veins (PV) –

oxygenated blood from the lungs to the heart

PULMONARY CIRCUIT





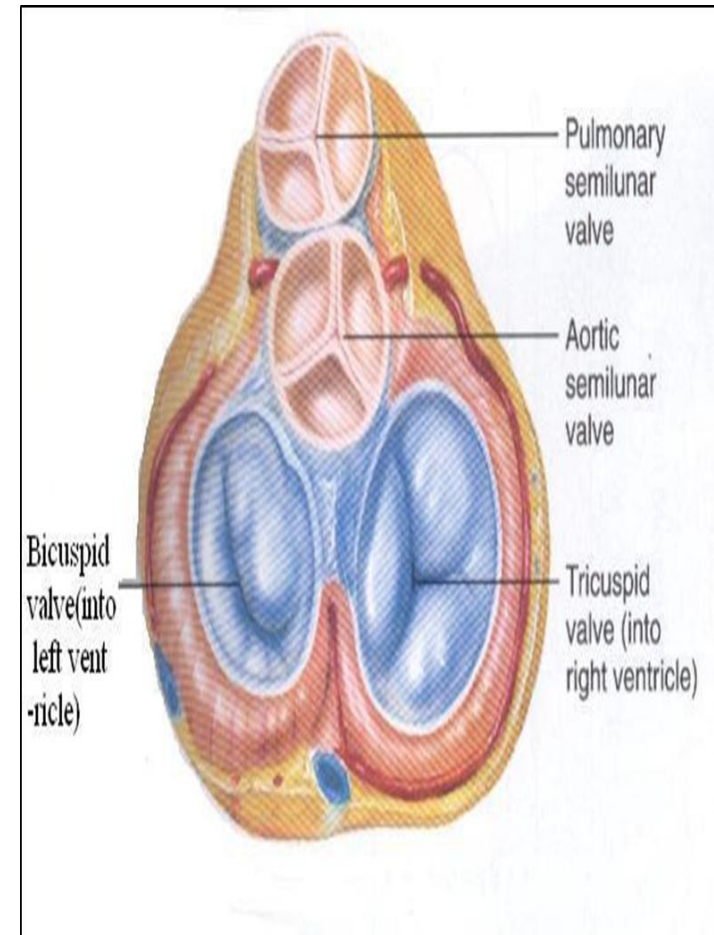
Atrioventricular (AV) valves:

Right side → Tricuspid valve (between right atrium and right ventricle)

Left side → Mitral valve (between left atrium and left ventricle)

Function of AV valves (mitral & tricuspid):

Allow one-way flow and Prevent backflow (regurgitation/leakage) of blood into the atria during ventricular contraction

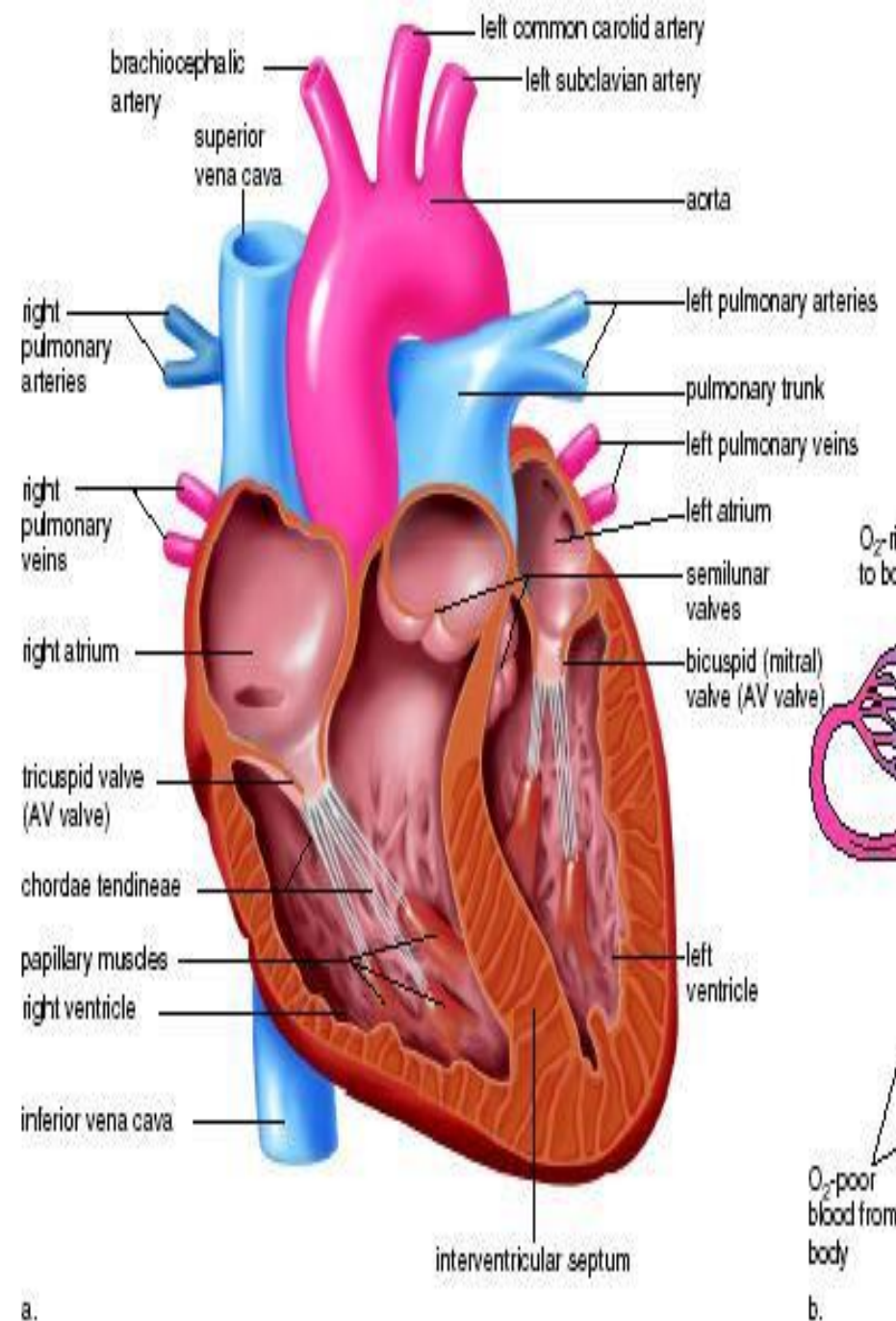


Atrioventricular (AV) valves:

Right side → Tricuspid valve (between right atrium and right ventricle)

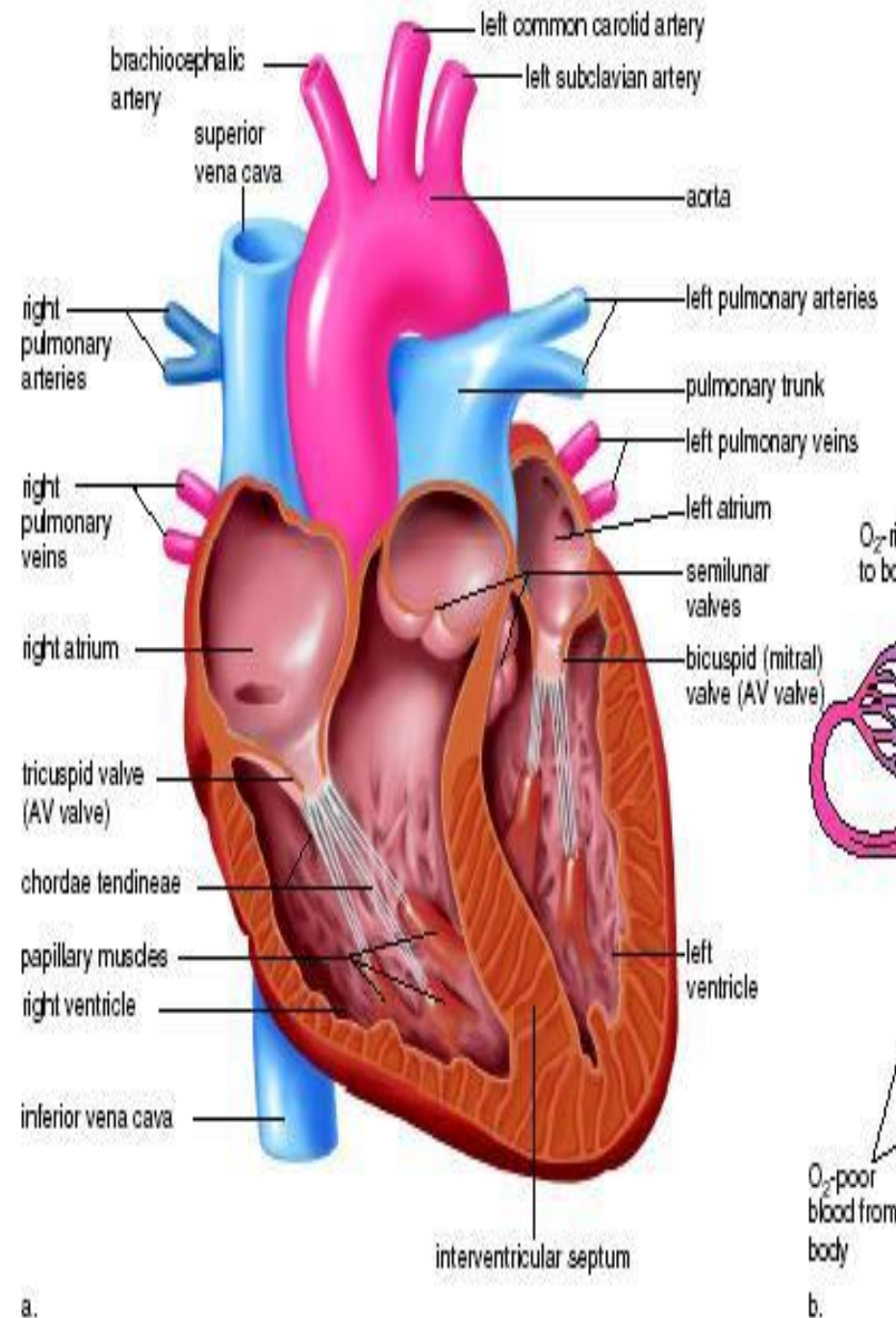
Left side → Mitral valve (between left atrium and left ventricle)

Allow one-way flow and Prevent backflow (regurgitation/leakage) of blood into the atria during ventricular contraction



Support structures:

- ❖ Papillary muscles (small muscular projections from ventricular walls)
- ❖ Chordae tendineae (“heart strings”) connect papillary muscles to the valve leaflets
- ❖ Papillary muscles contract along with the ventricular wall, This action tightens the chordae tendineae and Prevents eversion/prolapse of valve cusps back into the atria



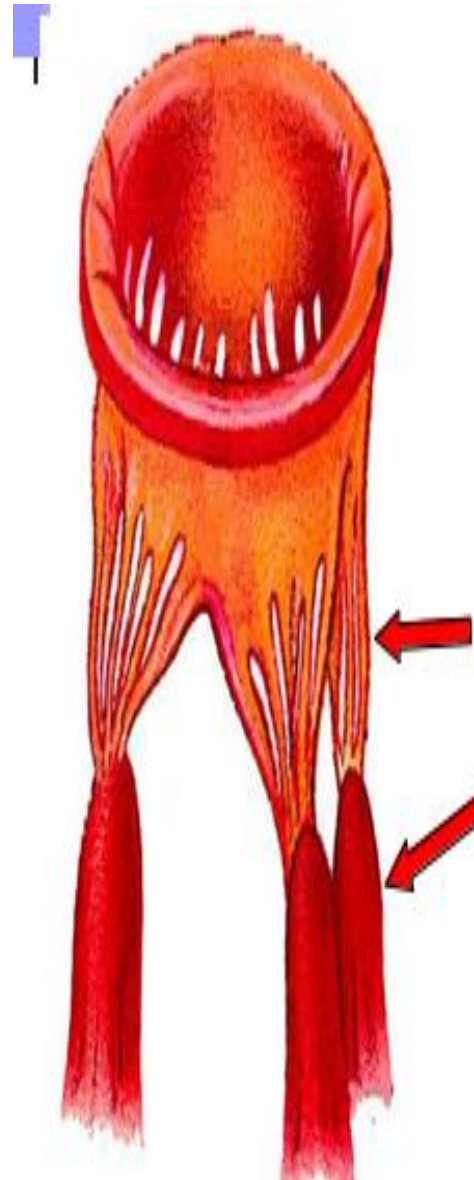
Importance:

Ensures one-way blood flow
(atria → ventricles → great arteries)

Prevents regurgitation of
blood into the atria despite
high ventricular pressure

Clinical relevance:

Papillary muscle dysfunction
(e.g., ischemia, rupture after
myocardial infarction) can
cause valvular insufficiency
→ backflow (regurgitation)
and heart failure symptoms.



- Dense connective tissue covered by endocardium
- AV valves
 - chordae tendineae - thin fibrous cords
 - connect valves to papillary muscles

Valve Function

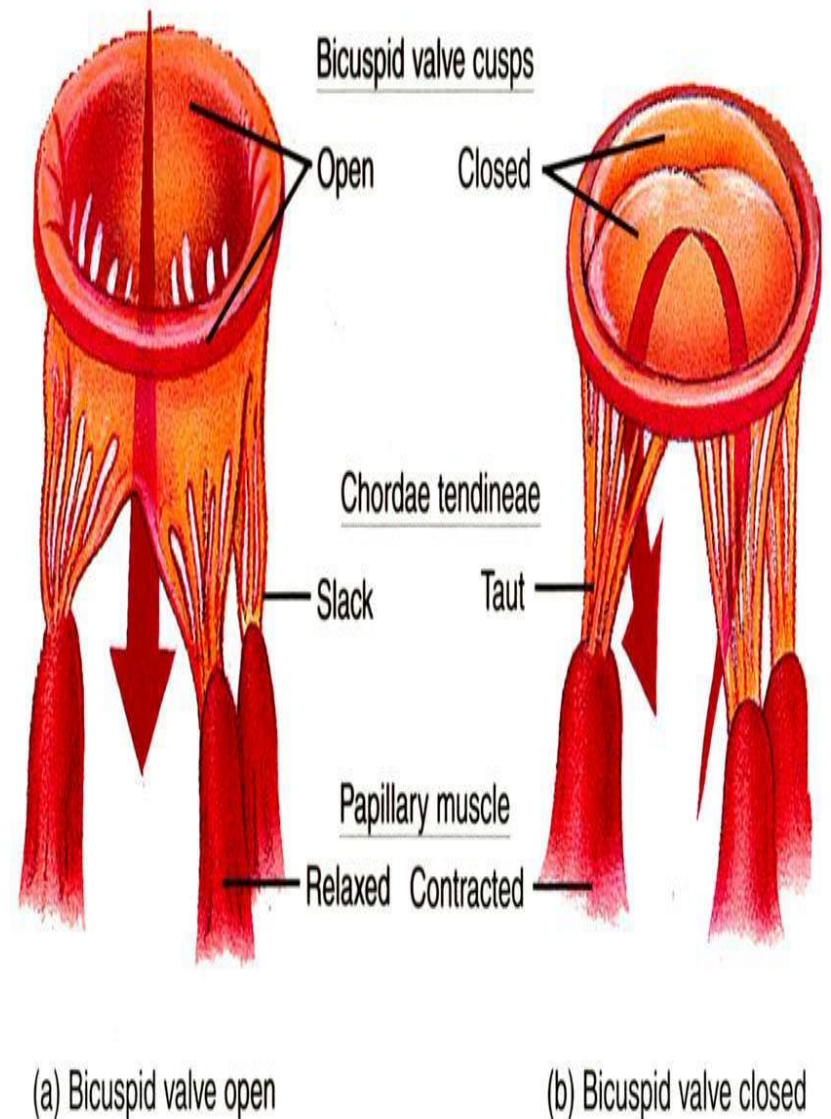
Mechanism of opening/closing
(passive process):

Open → when a forward pressure gradient (atria > ventricles) pushes blood forward

Close → when a backward pressure gradient (ventricles > atria) pushes blood backward

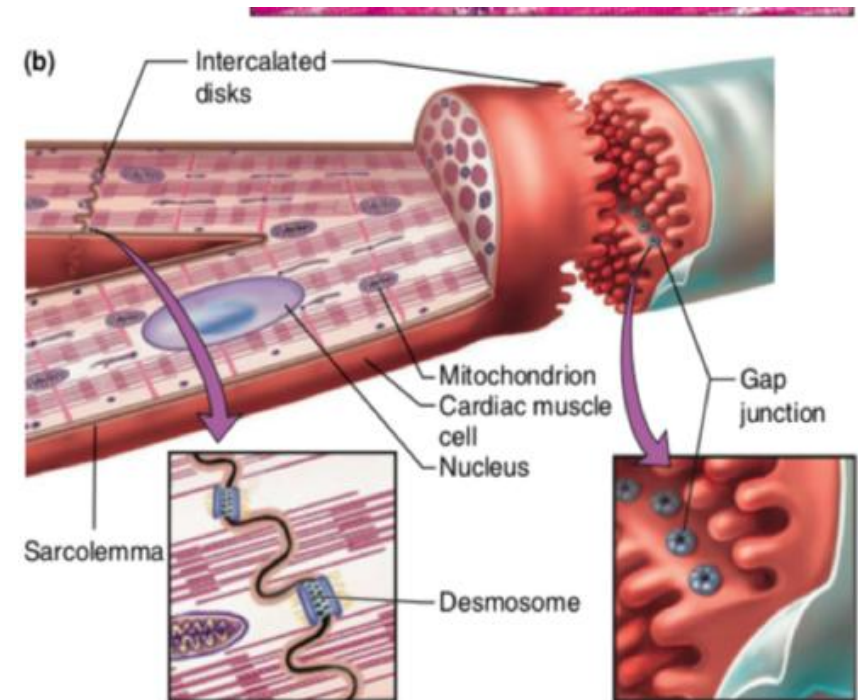
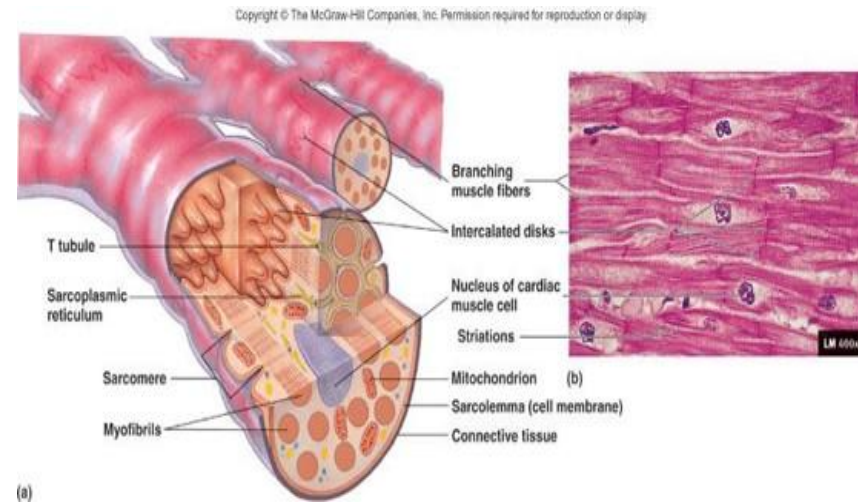
Key point:

AV valves themselves do not actively contract; they respond passively to pressure gradients.



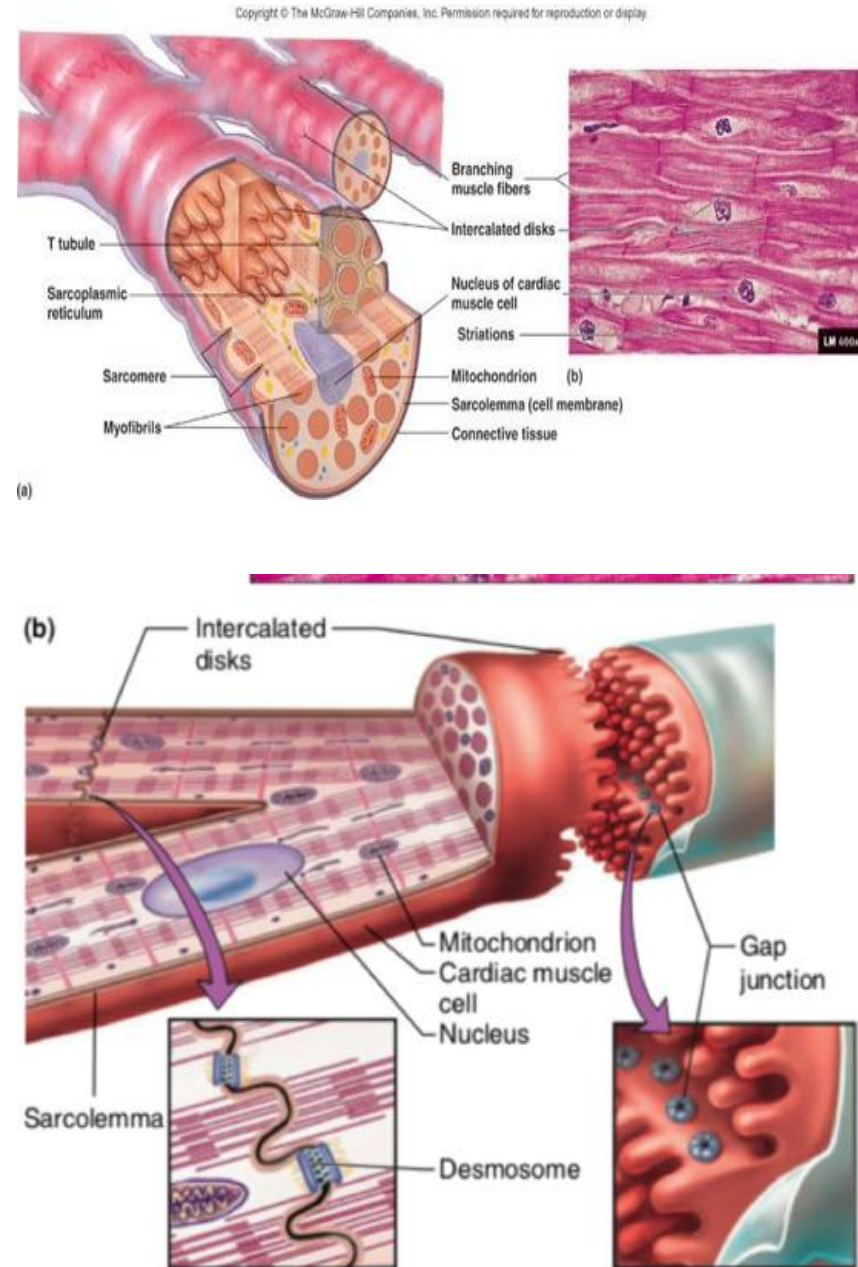
Cardiac Muscle

- ❖ The heart is composed of three major types of cardiac muscle
 - Atrial muscle
 - Ventricular muscle
 - Specialized excitatory and conductive fibers
- ❖ Atrial & ventricular muscle fibers contract similarly to skeletal muscle, but the contraction duration is much longer than in skeletal muscle
- ❖ the cardiac muscle arranged in a latticework, with the fibers dividing, recombining, and then spreading again.
- ❖ cardiac muscle is striated in the same manner as in skeletal muscle.



AP|R **Figure 0 30** Cardiac muscle. (a) Light micrograph. (b) Cardiac mus... Skaermbillede... Intercalated disks.

- ❖ Cardiac muscle has typical myofibrils that contain actin and myosin filaments almost identical to those found in skeletal muscle;
- ❖ these filaments lie side by side and slide during contraction in the same manner as occurs in skeletal muscle



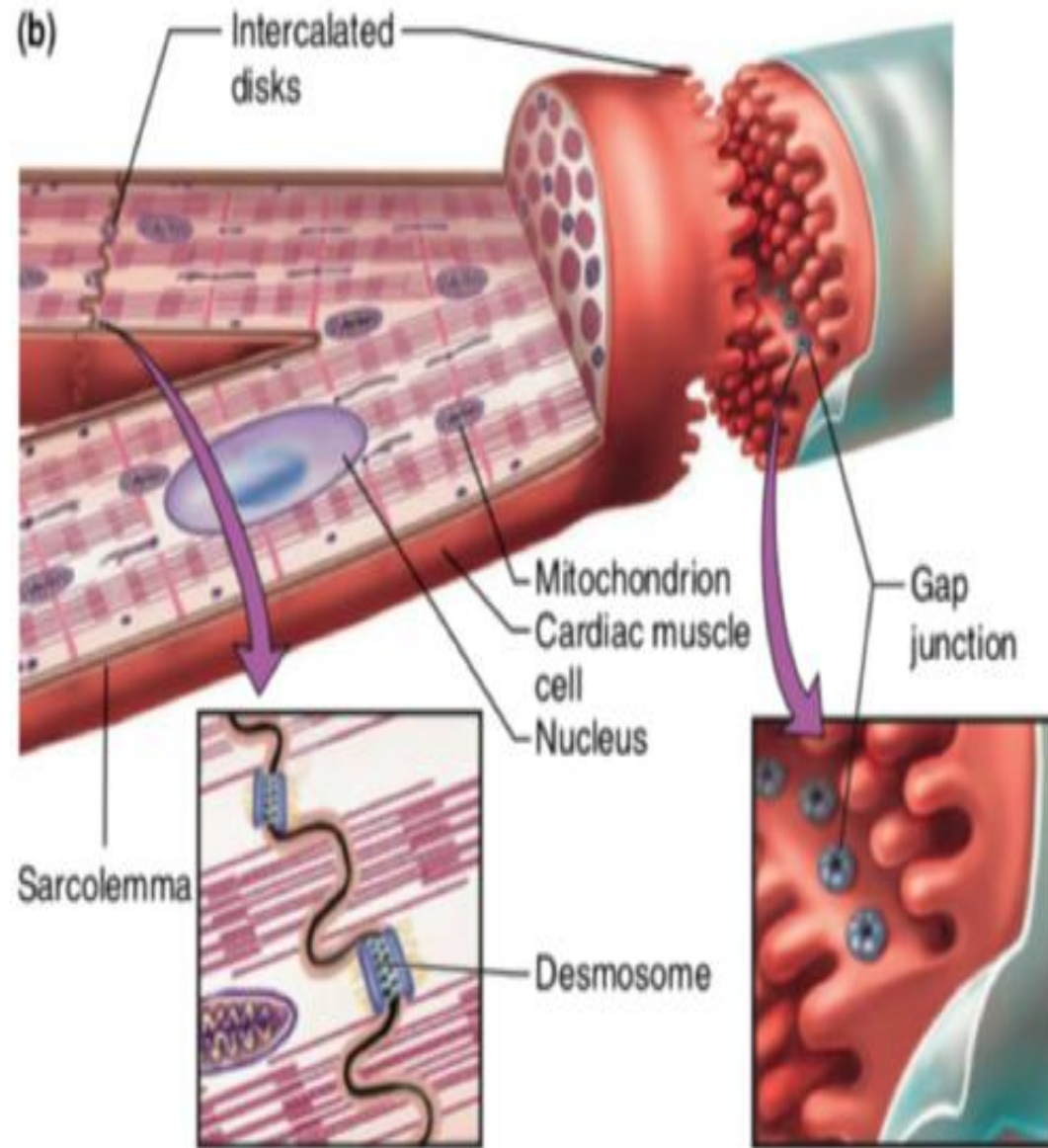
AP|R **Figure 0 30** Cardiac muscle. (a) Light micrograph. (b) Cardiac muscle showing intercalated disks.

❖ **Intercalated disks:**
specialized cell-cell contacts.

- Cell membranes interdigitate
- Desmosomes hold cells together
- Gap junctions allow action potentials to move from one cell to the next.

❖ Electrically, cardiac muscle of the atria and of the ventricles behaves as single unit

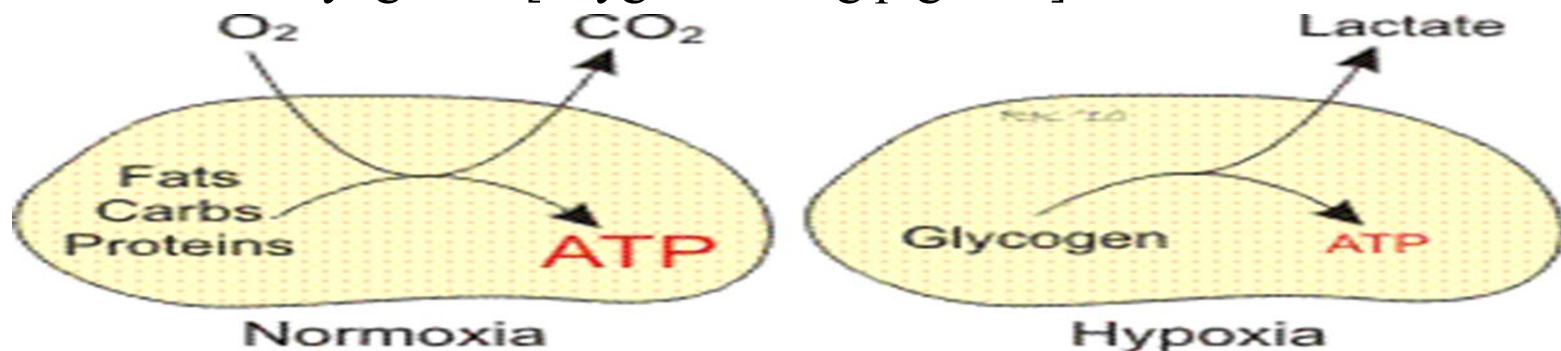
❖ Mitochondria comprise 30% of volume of the cell vs. 2% in skeletal muscle



APR **Figure 9.30** Cardiac muscle. (a) Light micrograph. (b) Cardiac muscle. Skaermbillede Intercalated disks.

Cardiac muscle does not undergo fatigue

- ❖ The heart can use a variety of substrates for oxidation ,
- ❖ The primary ATP source for contraction is produced by mitochondria by aerobic respiration.
- ❖ When a region of heart muscle is oxygen-starved (as during a heart attack), the ischemic cells (ischemic = blood deprived) begin to metabolize anaerobically, producing lactic acid. the heart is able to utilize glycogen (a storage form of carbohydrate) as a substrate for anaerobic production of ATP and the formation of lactic acid.
- ❖ Cardiac muscle does not undergo fatigue and this phenomenon has 3 reasons :
 - 1-Rich blood supply (oxygen-rich tissue).
 - 2-High numbers of mitochondriaEfficient **waste removal** prevents buildup of fatigue products
 3. Presence of myoglobin [oxygen-storing pigment].



Physiological properties of the cardiac muscle:

Electrical properties.

1. **Syncytium properties:** A piece of the heart behave as single unite to electrical stimulation , rather than a group of isolated units as in skeletal muscle. Myocardial fibers have ‘functional syncytium’ and not ‘anatomical syncytium

2. **Rhythmicity (Automaticity/Chronotropism)**

automaticity is the ability of the cardiac muscles to depolarize spontaneously, without external electrical stimulation from the nervous system

3. **Excitability:**

Is the ability of cardiac muscle to respond to adequate stimuli by generating an action potential.

4. **Conductivity (Dromotropism):**

Is the ability of cardiac muscle fibers to conduct the cardiac impulses that are initiated in the SA node.

Mechanical properties of cardiac muscle

1. Contractility(Inotropism)

It is the ability of the cardiac muscle to convert electrical energy into mechanical work

2. All-or-None Law of Cardiac Muscle

- Sub-threshold stimulus → No response
- Threshold stimulus → Maximal contraction
- Stronger stimuli → No further increase in contraction

3. Long absolute refractory period:

Cardiac muscle cannot be stimulated while it is contracted so it cannot be tetanized.

4. Plateau phase :long action potential & sustained contraction:

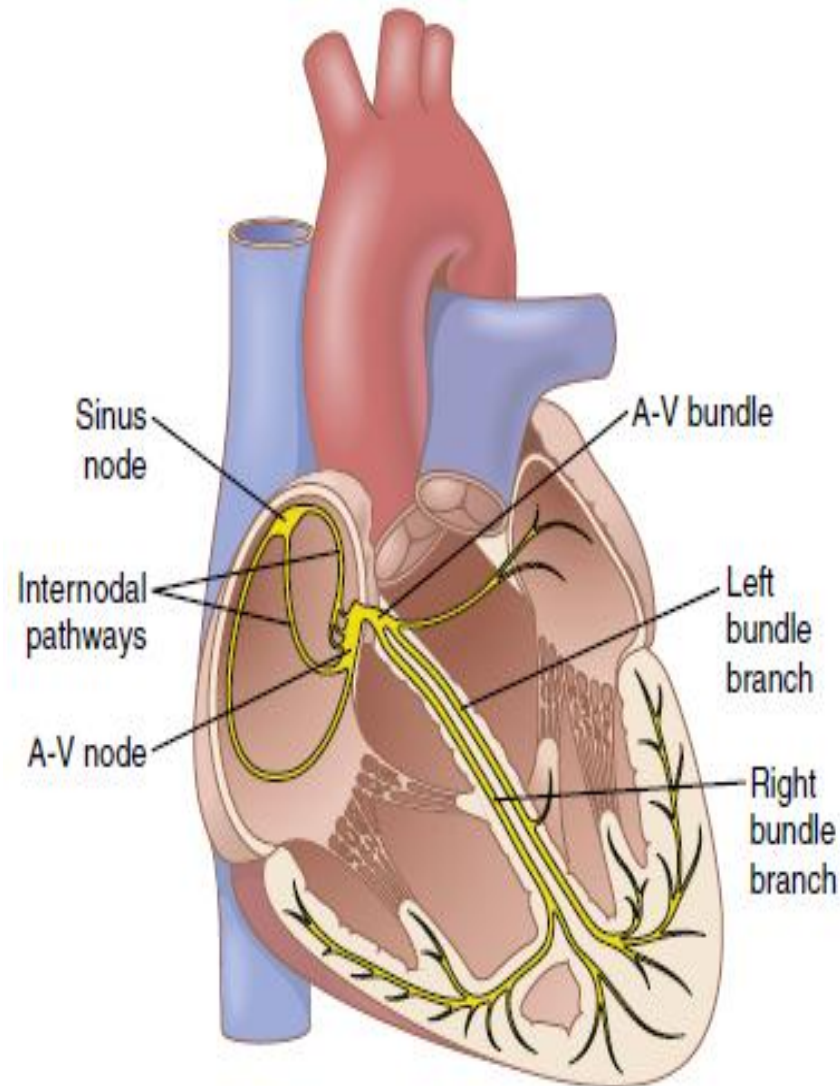
It is a unique feature of cardiac muscle (not seen in skeletal muscle).

NOTE :

- In skeletal muscle , Repeated stimuli can arrive before the muscle has relaxed, leading to summation and then tetanus
- but in the heart , plateau phase + long refractory period prevent tetanus and allow rhythmical contractions

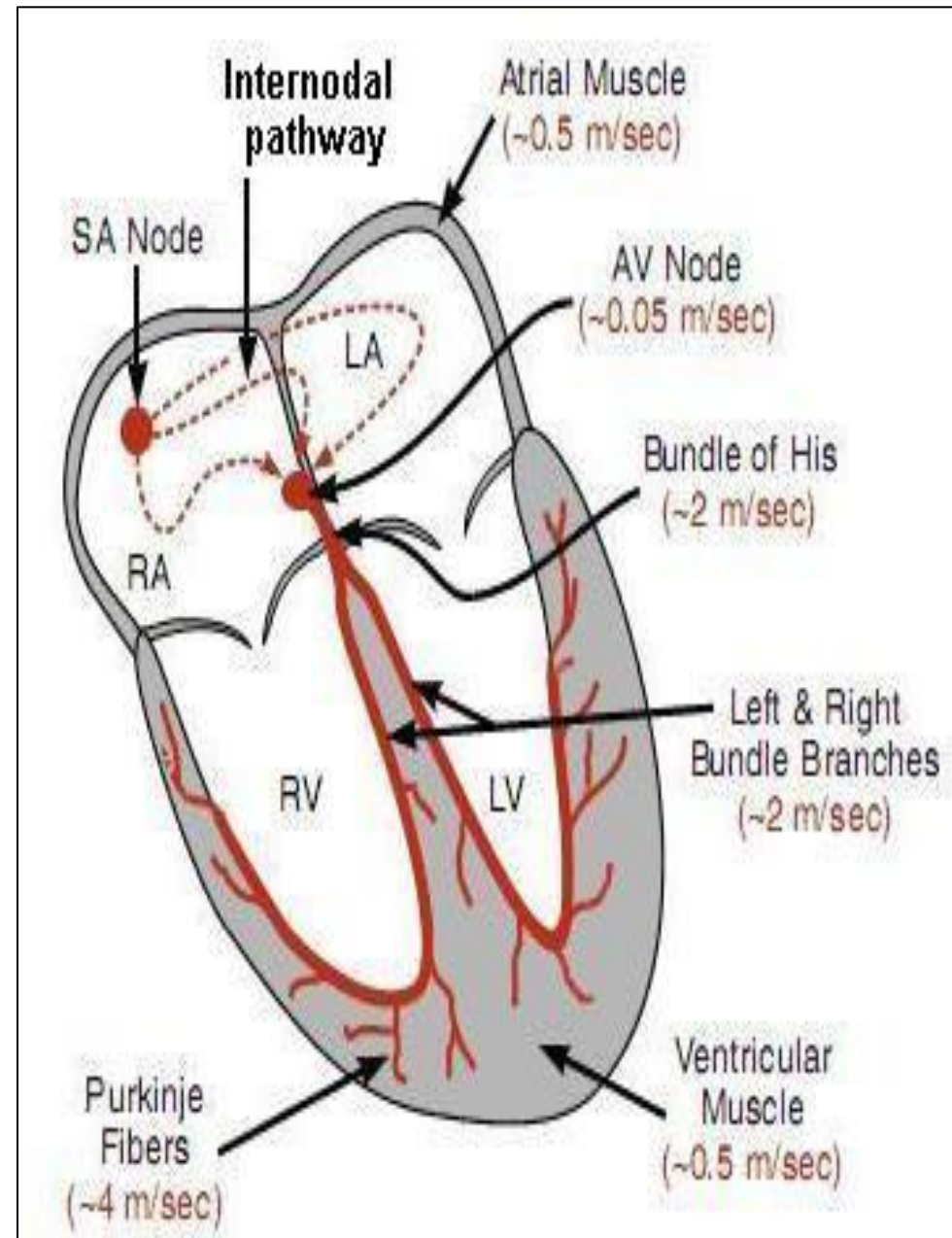
Conductive system of the heart

- ❖ Made of specialized myocardial fibers.
- ❖ Contract weakly → contain few contractile fibrils.
- ❖ Main function is not contraction, but:
 - Automatic rhythmic discharge (generation of action potentials i.e. pacemaker).
 - Conduction of impulses throughout the heart.
- ❖ It consist of :
 1. Sino-atrial (SA) node
 2. Atrio-ventricular (AV) node
 3. Bundle of His : It is the continuation of the AV node and is located in the interventricular septum
 4. Bundle branches
 5. Purkinje fibers



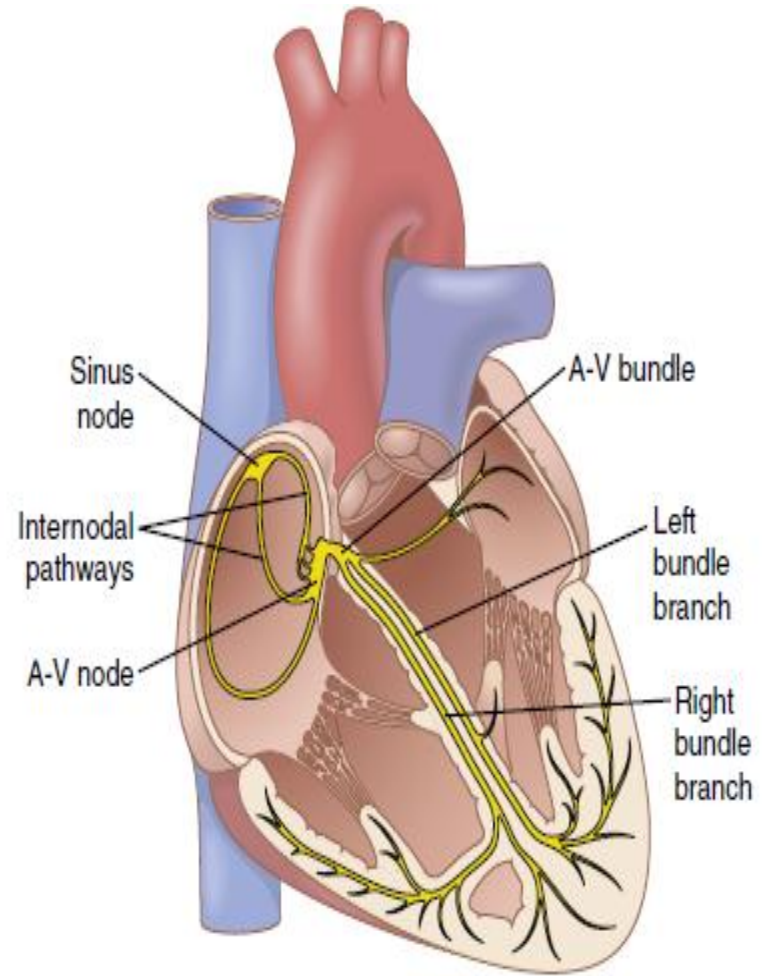
The conduction velocity of the conduction system of the heart.

Tissue	Resting membrane potential (mV)	Conduction velocity (m/sec)
SA node	40-60	0.05
Atrial muscle	70-80	0.5
Internodal tracts	80-90	1.0
AV node	60	0.05
Purkinje fibers	70	4.0
Ventricular muscle	90	0.5



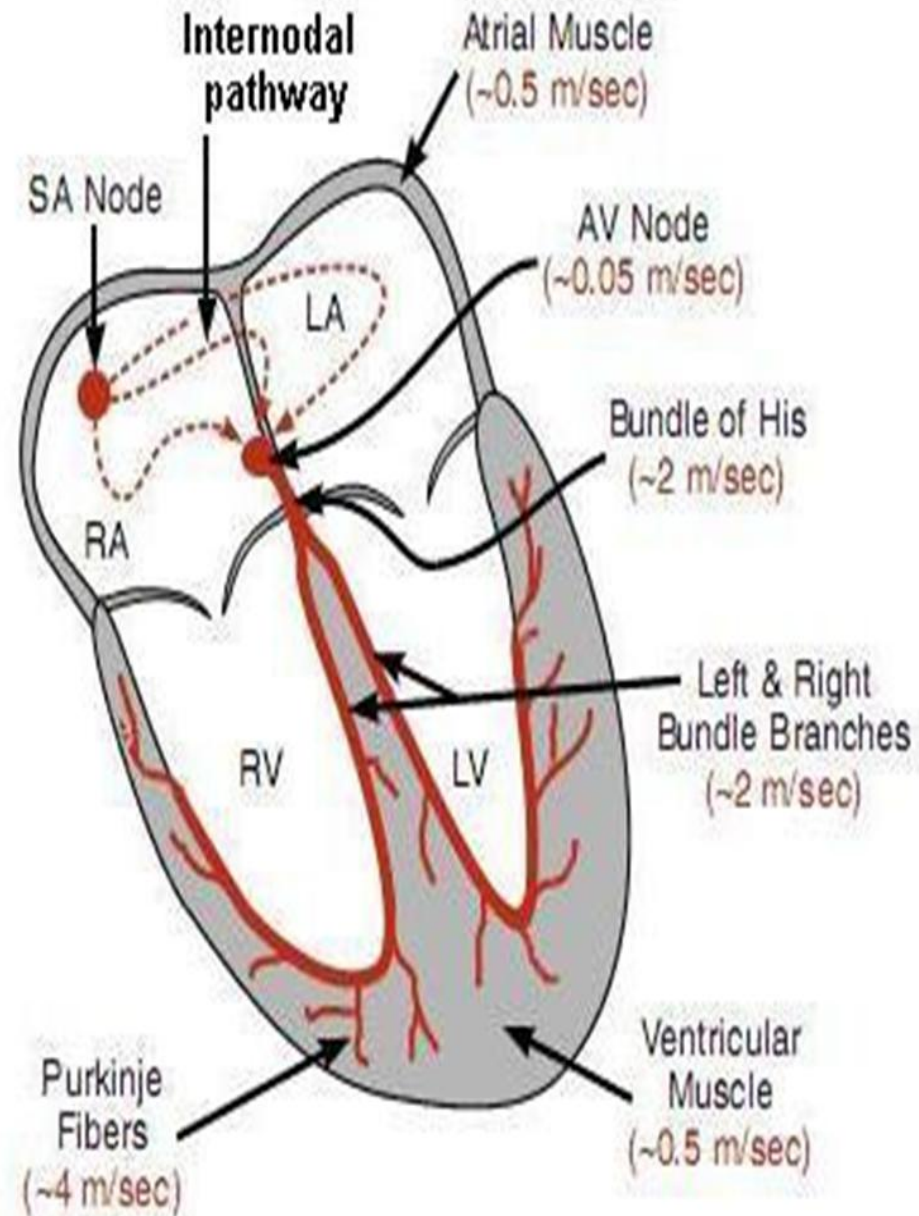
[1] S.A NODE:

- ❖ It is located near the junction of the superior vena cava and the right atrium
- ❖ Their fibers are continuous with the atrial fibers so that any action potential that begins in the SA node spreads immediately to the atria.
- ❖ It is the normal primary pacemaker of the heart in which rhythmic self-excitatory impulse is generated (it has the property of automaticity i.e unstable resting membrane potential .



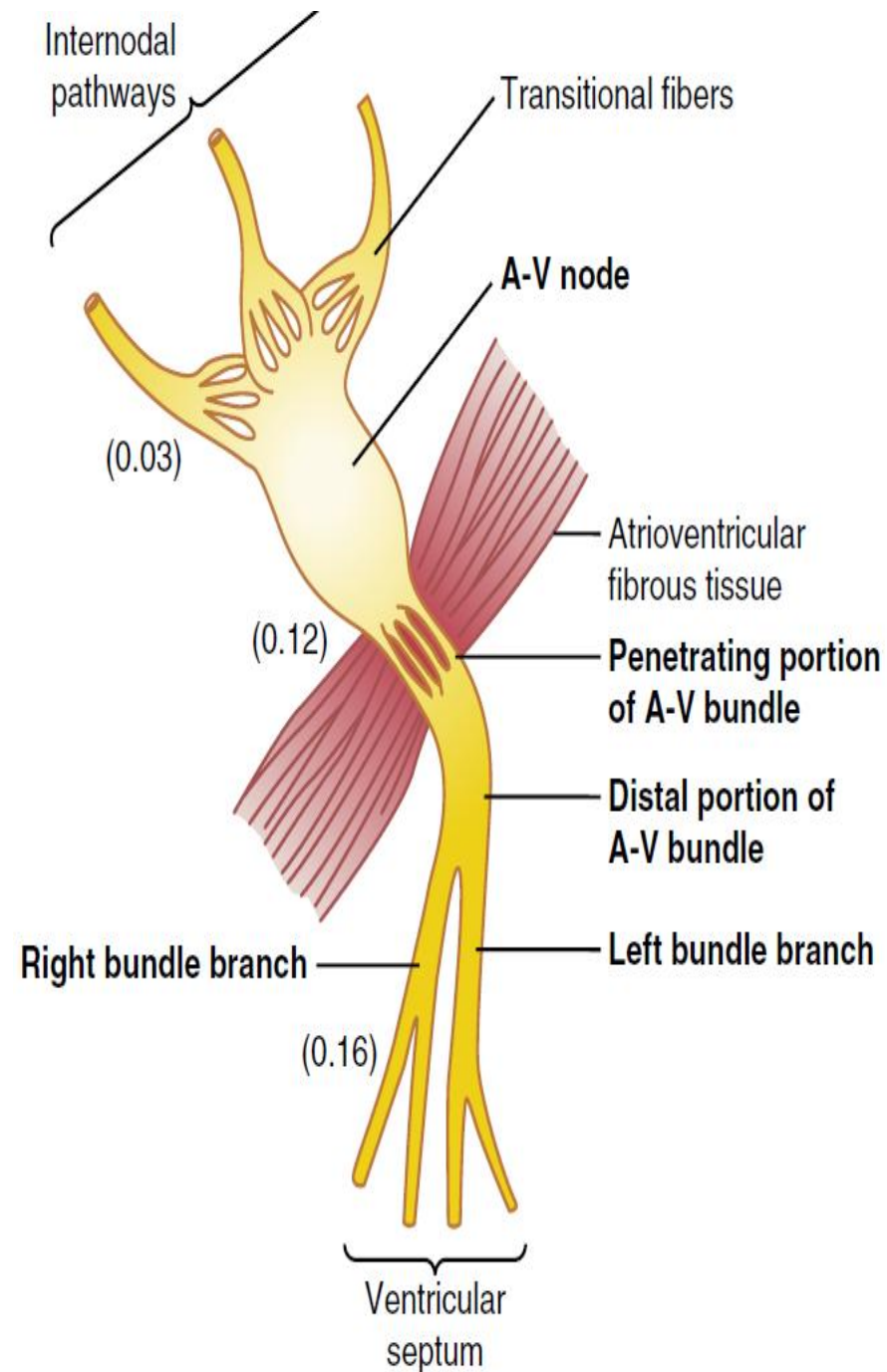
[2] Interatrial tract (Bachmann's bundle): This band of specialized muscle fibers runs from the SA node to the left atrium and causes almost simultaneous depolarization and contraction of both atria because the conduction velocity through atrial muscle is very fast [0.5m / second].

[3] Internodal tract: Three bundles of specialized cells (anterior, middle, and posterior) connect the SA and AV nodes. The internodal tracts increase the likelihood that impulses from SA node will reach the AV node and initiate ventricular depolarization.



[4] AV (atrioventricular) node:

- Located at the Posterior wall of the **right atrium**, just behind the **tricuspid valve**.
- The AV node is normally the only path for excitation to proceed from the atria to the ventricles.
- The AV node **delays conduction** of impulses from atria → ventricles.
- SA node → AV node: 0.03 sec
- Delay within AV node: 0.09 sec
- Delay in penetrating AV bundle: 0.04 sec
- Total delay = ~0.16 sec before excitation reaches ventricular muscle.



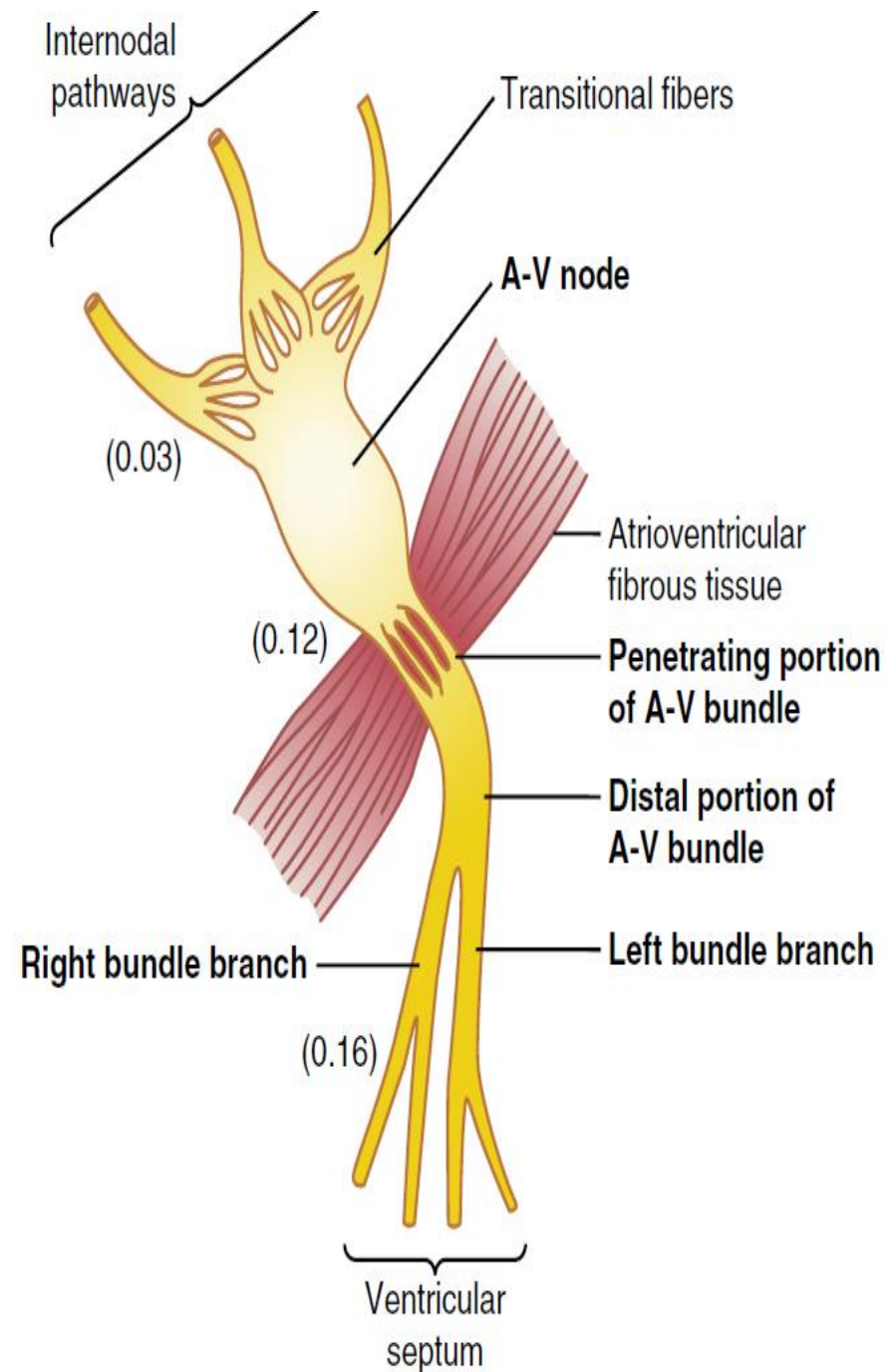
- This delay provides time for atrial contraction to occur, in which the atria contract before the ventricles, which enhances ventricular filling.

Cause of Slow Conduction

AV nodal and bundle fibers have **fewer gap junctions**.

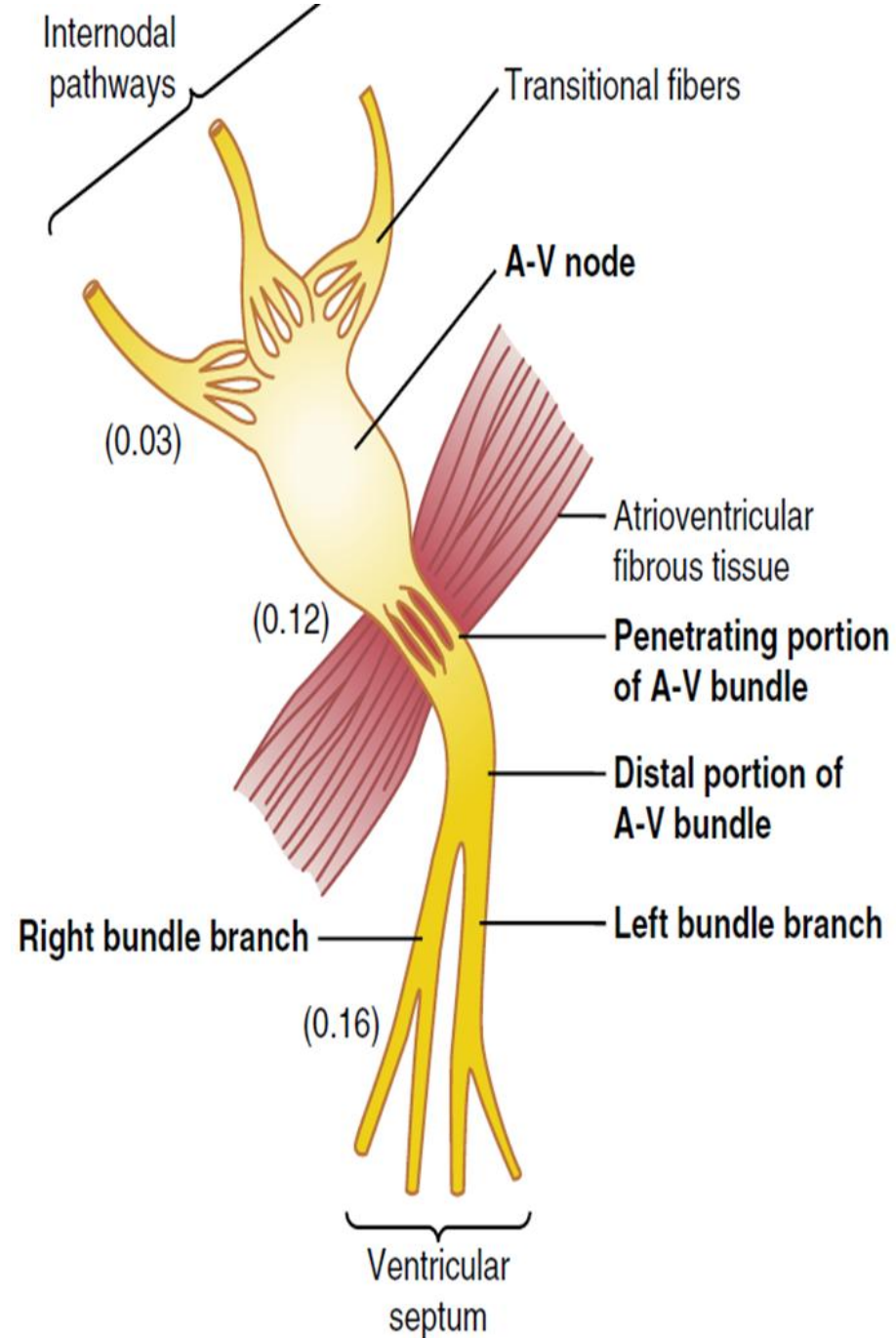
This increases **resistance to ion flow** → slows impulse conduction.

AV node acts as a **physiological gatekeeper**—slowing conduction to coordinate atrial systole with ventricular systole.



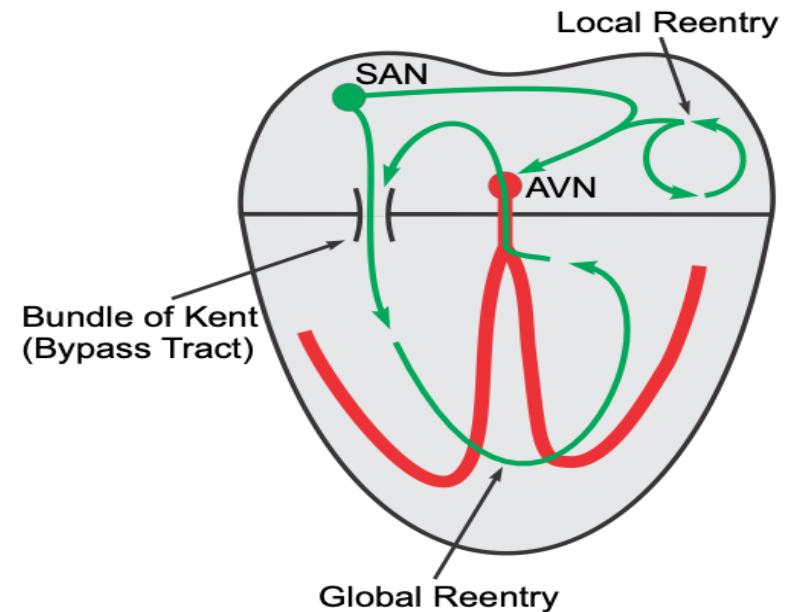
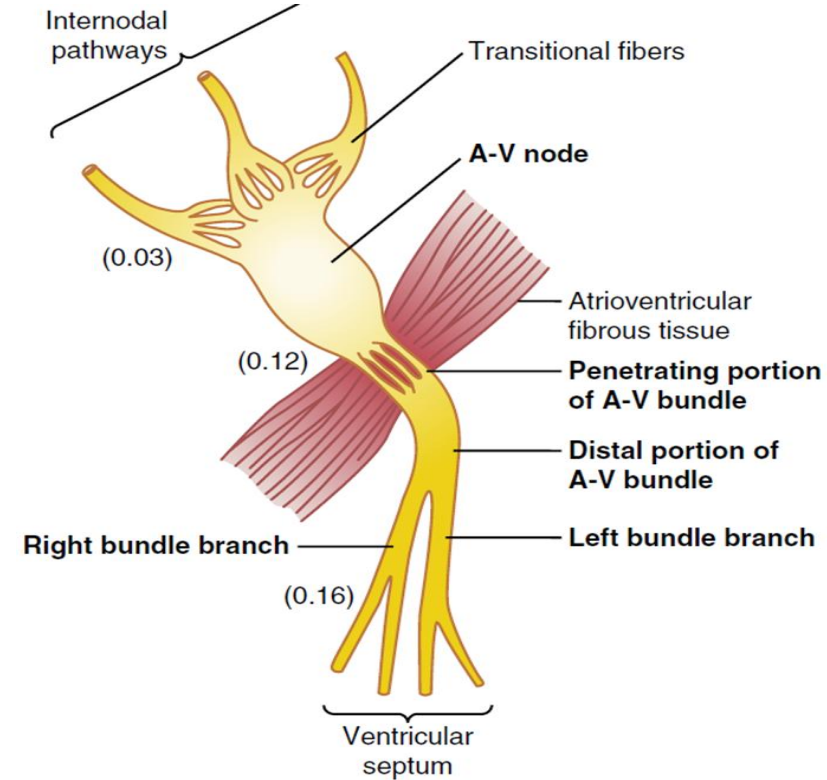
[5] Bundle of His (AV Bundle):

- ❖ It is the continuation of the AV node and is located in the interventricular septum.
- ❖ The bundle of His divides into the right and left bundle branches, then left bundle branch divided into anterior & posterior fascicles.
- ❖ Atria and ventricles are separated by a continuous fibrous skeleton which acts as an insulator, blocking conduction except through the AV bundle.



[5] Bundle of His (AV Bundle):

- ❖ AV bundle conducts impulses only from atria → ventricles.
- ❖ Prevents backward conduction (ventricles → atria) under normal conditions.
- ❖ In rare cases, an **accessory pathway** penetrates the fibrous barrier outside the AV bundle, allowing impulses to re-enter the atria from the ventricles and potentially causing **serious re-entry arrhythmias**



[6] The right and left bundle branches

proceed on each side of the interventricular septum to their respective ventricles.

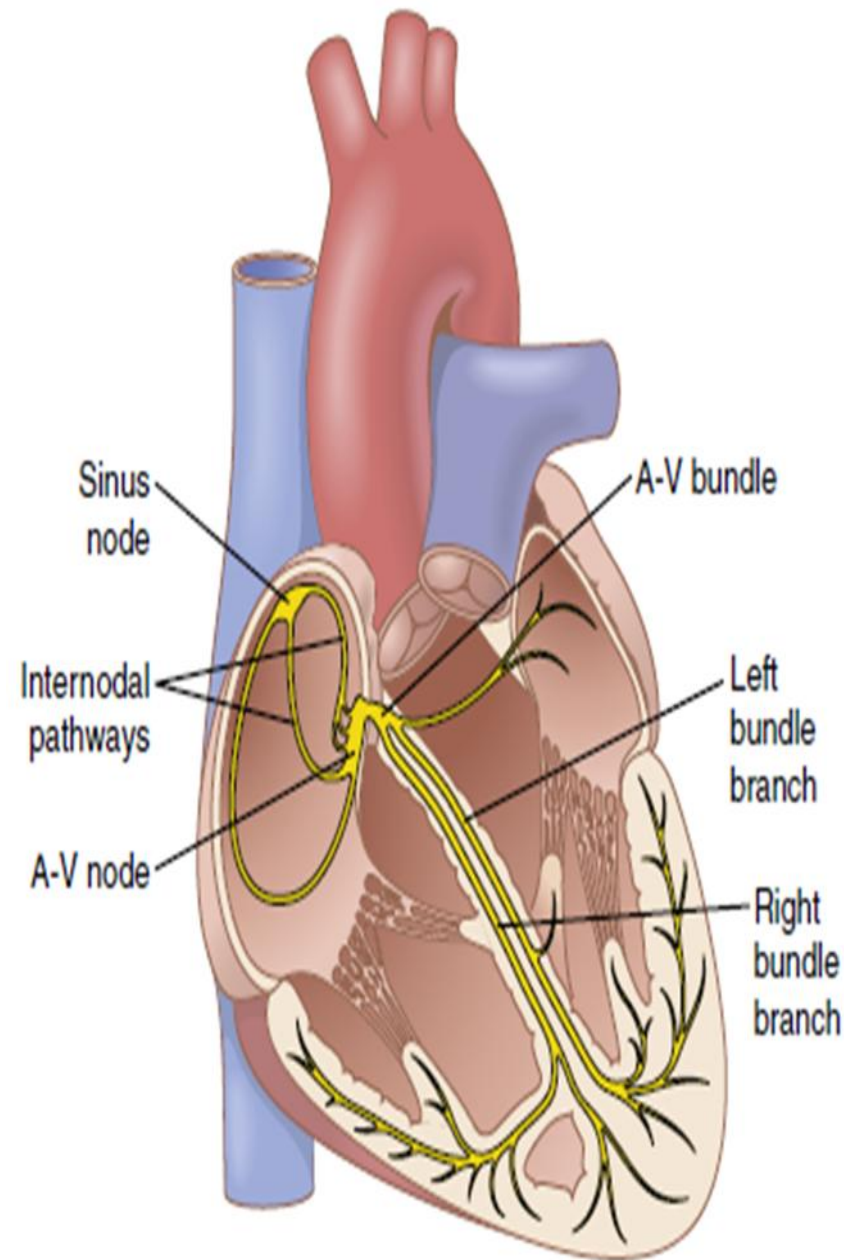
[7] Purkinje fibers

Purkinje fibers arise from the **AV bundle** and spread into ventricles.

Very large fibers → larger than ventricular muscle fibers.

Conduction Velocity in these fibers is 1.5–4.0 m/s ($\approx 6\times$ faster than ventricular muscle, $\approx 150\times$ faster than AV nodal fibers).

This Ensures **near-simultaneous activation** of the entire ventricular myocardium.



- ❖ **The Fast Conduction in Purkinje fibers is due to Abundant gap junctions** at intercalated discs → low resistance to ion flow between cells → high conduction velocity.
- ❖ Since the electrical stimulus begins at the apex, the contraction also begins at the apex and travels toward the base of the heart.
- ❖ This allows the blood to be pumped out of the ventricles and into the aorta and pulmonary trunk

